```python
# Recurrent Neural Network



# Part 1 - Data Preprocessing

# Importing the libraries
import numpy as np
import matplotlib.pyplot as plt
import pandas as pd

# Importing the training set
dataset_train = pd.read_csv('/content/Google_Stock_Price_Train.csv')
training_set = dataset_train.iloc[:, 1:2].values

# Feature Scaling
from sklearn.preprocessing import MinMaxScaler
sc = MinMaxScaler(feature_range = (0, 1))
training_set_scaled = sc.fit_transform(training_set)

# Creating a data structure with 60 timesteps and 1 output
X_train = []
y_train = []
for i in range(60, 1258):
    X_train.append(training_set_scaled[i-60:i, 0])
    y_train.append(training_set_scaled[i, 0])
X_train, y_train = np.array(X_train), np.array(y_train)

# Reshaping
X_train = np.reshape(X_train, (X_train.shape[0], X_train.shape[1], 1))

# Part 2 - Building the RNN


# Importing the Keras libraries and packages
from keras.models import Sequential
from keras.layers import Dense
from keras.layers import LSTM
from keras.layers import Dropout
from keras.layers import Input

# Initialising the RNN
regressor = Sequential()

# Adding the first LSTM layer and some Dropout regularisation
timesteps = 60
input_dim = 1
regressor.add(Input(shape=(timesteps, input_dim)))
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regressor.add(LSTM(units = 50, return_sequences = True))
regressor.add(Dropout(0.2))

# Adding the second LSTM layer and some Dropout regularisation
regressor.add(LSTM(units = 50, return_sequences = True))
regressor.add(Dropout(0.2))

# Adding the third LSTM layer and some Dropout regularisation
regressor.add(LSTM(units = 50, return_sequences = True))
regressor.add(Dropout(0.2))

# Adding the fourth LSTM layer and some Dropout regularisation
regressor.add(LSTM(units = 50))
regressor.add(Dropout(0.2))

# Adding the output layer
regressor.add(Dense(units = 1))

#Compiling the RNN
regressor.compile(optimizer = 'adam', loss = 'mean_squared_error')

#Fitting the RNN to the Training set
regressor.fit(X_train, y_train, epochs = 100, batch_size = 32)
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Epoch 1/100
38/38 ━━━━━━━━━━━━━━━━━━━━ 12s 105ms/step - loss: 0.1114
Epoch 2/100
38/38 ━━━━━━━━━━━━━━━━━━━━ 4s 105ms/step - loss: 0.0077
Epoch 3/100
38/38 ━━━━━━━━━━━━━━━━━━━━ 6s 142ms/step - loss: 0.0056
Epoch 4/100
38/38 ━━━━━━━━━━━━━━━━━━━━ 9s 105ms/step - loss: 0.0045
Epoch 5/100
38/38 ━━━━━━━━━━━━━━━━━━━━ 6s 136ms/step - loss: 0.0048
Epoch 6/100
38/38 ━━━━━━━━━━━━━━━━━━━━ 10s 122ms/step - loss: 0.0050
Epoch 7/100
38/38 ━━━━━━━━━━━━━━━━━━━━ 5s 116ms/step - loss: 0.0050
Epoch 8/100
38/38 ━━━━━━━━━━━━━━━━━━━━ 5s 106ms/step - loss: 0.0043
Epoch 9/100
38/38 ━━━━━━━━━━━━━━━━━━━━ 7s 144ms/step - loss: 0.0043
Epoch 10/100
38/38 ━━━━━━━━━━━━━━━━━━━━ 9s 105ms/step - loss: 0.0040
Epoch 11/100
38/38 ━━━━━━━━━━━━━━━━━━━━ 6s 133ms/step - loss: 0.0043
Epoch 12/100
38/38 ━━━━━━━━━━━━━━━━━━━━ 10s 130ms/step - loss: 0.0050
Epoch 13/100
38/38 ━━━━━━━━━━━━━━━━━━━━ 5s 121ms/step - loss: 0.0040
Epoch 14/100
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38/38 ━━━━━━━━━━━━━━━━━━━━ 4s 105ms/step - loss: 0.0044
Epoch 15/100
38/38 ━━━━━━━━━━━━━━━━━━━━ 5s 132ms/step - loss: 0.0034
Epoch 16/100
38/38 ━━━━━━━━━━━━━━━━━━━━ 4s 112ms/step - loss: 0.0036
Epoch 17/100
38/38 ━━━━━━━━━━━━━━━━━━━━ 4s 108ms/step - loss: 0.0035
Epoch 18/100
38/38 ━━━━━━━━━━━━━━━━━━━━ 5s 139ms/step - loss: 0.0031
Epoch 19/100
38/38 ━━━━━━━━━━━━━━━━━━━━ 9s 105ms/step - loss: 0.0033
Epoch 20/100
38/38 ━━━━━━━━━━━━━━━━━━━━ 6s 141ms/step - loss: 0.0036
Epoch 21/100
38/38 ━━━━━━━━━━━━━━━━━━━━ 9s 110ms/step - loss: 0.0038
Epoch 22/100
38/38 ━━━━━━━━━━━━━━━━━━━━ 6s 122ms/step - loss: 0.0032
Epoch 23/100
38/38 ━━━━━━━━━━━━━━━━━━━━ 4s 104ms/step - loss: 0.0036
Epoch 24/100
38/38 ━━━━━━━━━━━━━━━━━━━━ 7s 145ms/step - loss: 0.0033
Epoch 25/100
38/38 ━━━━━━━━━━━━━━━━━━━━ 9s 106ms/step - loss: 0.0033
Epoch 26/100
38/38 ━━━━━━━━━━━━━━━━━━━━ 5s 143ms/step - loss: 0.0031
Epoch 27/100
38/38 ━━━━━━━━━━━━━━━━━━━━ 4s 104ms/step - loss: 0.0033
Epoch 28/100
38/38 ━━━━━━━━━━━━━━━━━━━━ 5s 107ms/step - loss: 0.0028
Epoch 29/100
38/38 ━━━━━━━━━━━━━━━━━━━━ 6s 124ms/step - loss: 0.0034
Epoch 30/100
38/38 ━━━━━━━━━━━━━━━━━━━━ 4s 105ms/step - loss: 0.0026
Epoch 31/100
38/38 ━━━━━━━━━━━━━━━━━━━━ 5s 121ms/step - loss: 0.0031
Epoch 32/100
38/38 ━━━━━━━━━━━━━━━━━━━━ 5s 115ms/step - loss: 0.0030
Epoch 33/100
38/38 ━━━━━━━━━━━━━━━━━━━━ 4s 106ms/step - loss: 0.0028
Epoch 34/100
38/38 ━━━━━━━━━━━━━━━━━━━━ 5s 125ms/step - loss: 0.0028
Epoch 35/100
38/38 ━━━━━━━━━━━━━━━━━━━━ 5s 116ms/step - loss: 0.0029
Epoch 36/100
38/38 ━━━━━━━━━━━━━━━━━━━━ 4s 106ms/step - loss: 0.0031
Epoch 37/100
38/38 ━━━━━━━━━━━━━━━━━━━━ 6s 144ms/step - loss: 0.0025
Epoch 38/100
38/38 ━━━━━━━━━━━━━━━━━━━━ 9s 104ms/step - loss: 0.0024
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Epoch 39/100
38/38 ━━━━━━━━━━━━━━━━━━━━ 7s 144ms/step - loss: 0.0031
Epoch 40/100
38/38 ━━━━━━━━━━━━━━━━━━━━ 9s 109ms/step - loss: 0.0026
Epoch 41/100
38/38 ━━━━━━━━━━━━━━━━━━━━ 6s 121ms/step - loss: 0.0024
Epoch 42/100
38/38 ━━━━━━━━━━━━━━━━━━━━ 4s 104ms/step - loss: 0.0025
Epoch 43/100
38/38 ━━━━━━━━━━━━━━━━━━━━ 5s 124ms/step - loss: 0.0022
Epoch 44/100
38/38 ━━━━━━━━━━━━━━━━━━━━ 5s 123ms/step - loss: 0.0025
Epoch 45/100
38/38 ━━━━━━━━━━━━━━━━━━━━ 4s 104ms/step - loss: 0.0025
Epoch 46/100
38/38 ━━━━━━━━━━━━━━━━━━━━ 5s 125ms/step - loss: 0.0025
Epoch 47/100
38/38 ━━━━━━━━━━━━━━━━━━━━ 5s 119ms/step - loss: 0.0024
Epoch 48/100
38/38 ━━━━━━━━━━━━━━━━━━━━ 4s 105ms/step - loss: 0.0029
Epoch 49/100
38/38 ━━━━━━━━━━━━━━━━━━━━ 5s 121ms/step - loss: 0.0024
Epoch 50/100
38/38 ━━━━━━━━━━━━━━━━━━━━ 5s 125ms/step - loss: 0.0022
Epoch 51/100
38/38 ━━━━━━━━━━━━━━━━━━━━ 4s 107ms/step - loss: 0.0023
Epoch 52/100
38/38 ━━━━━━━━━━━━━━━━━━━━ 6s 131ms/step - loss: 0.0022
Epoch 53/100
38/38 ━━━━━━━━━━━━━━━━━━━━ 4s 109ms/step - loss: 0.0028
Epoch 54/100
38/38 ━━━━━━━━━━━━━━━━━━━━ 4s 105ms/step - loss: 0.0022
Epoch 55/100
38/38 ━━━━━━━━━━━━━━━━━━━━ 7s 147ms/step - loss: 0.0021
Epoch 56/100
38/38 ━━━━━━━━━━━━━━━━━━━━ 9s 104ms/step - loss: 0.0025
Epoch 57/100
38/38 ━━━━━━━━━━━━━━━━━━━━ 5s 145ms/step - loss: 0.0025
Epoch 58/100
38/38 ━━━━━━━━━━━━━━━━━━━━ 4s 105ms/step - loss: 0.0022
Epoch 59/100
38/38 ━━━━━━━━━━━━━━━━━━━━ 4s 106ms/step - loss: 0.0022
Epoch 60/100
38/38 ━━━━━━━━━━━━━━━━━━━━ 7s 145ms/step - loss: 0.0022
Epoch 61/100
38/38 ━━━━━━━━━━━━━━━━━━━━ 9s 107ms/step - loss: 0.0019
Epoch 62/100
38/38 ━━━━━━━━━━━━━━━━━━━━ 5s 140ms/step - loss: 0.0021
Epoch 63/100
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38/38 ━━━━━━━━━━━━━━━━━━━━ 9s 110ms/step - loss: 0.0021
Epoch 64/100
38/38 ━━━━━━━━━━━━━━━━━━━━ 5s 118ms/step - loss: 0.0022
Epoch 65/100
38/38 ━━━━━━━━━━━━━━━━━━━━ 5s 105ms/step - loss: 0.0019
Epoch 66/100
38/38 ━━━━━━━━━━━━━━━━━━━━ 6s 143ms/step - loss: 0.0017
Epoch 67/100
38/38 ━━━━━━━━━━━━━━━━━━━━ 9s 104ms/step - loss: 0.0022
Epoch 68/100
38/38 ━━━━━━━━━━━━━━━━━━━━ 6s 137ms/step - loss: 0.0021
Epoch 69/100
38/38 ━━━━━━━━━━━━━━━━━━━━ 4s 105ms/step - loss: 0.0021
Epoch 70/100
38/38 ━━━━━━━━━━━━━━━━━━━━ 6s 119ms/step - loss: 0.0021
Epoch 71/100
38/38 ━━━━━━━━━━━━━━━━━━━━ 5s 118ms/step - loss: 0.0016
Epoch 72/100
38/38 ━━━━━━━━━━━━━━━━━━━━ 5s 103ms/step - loss: 0.0023
Epoch 73/100
38/38 ━━━━━━━━━━━━━━━━━━━━ 7s 143ms/step - loss: 0.0019
Epoch 74/100
38/38 ━━━━━━━━━━━━━━━━━━━━ 9s 105ms/step - loss: 0.0020
Epoch 75/100
38/38 ━━━━━━━━━━━━━━━━━━━━ 5s 141ms/step - loss: 0.0019
Epoch 76/100
38/38 ━━━━━━━━━━━━━━━━━━━━ 4s 106ms/step - loss: 0.0019
Epoch 77/100
38/38 ━━━━━━━━━━━━━━━━━━━━ 5s 110ms/step - loss: 0.0018
Epoch 78/100
38/38 ━━━━━━━━━━━━━━━━━━━━ 5s 137ms/step - loss: 0.0017
Epoch 79/100
38/38 ━━━━━━━━━━━━━━━━━━━━ 10s 123ms/step - loss: 0.0017
Epoch 80/100
38/38 ━━━━━━━━━━━━━━━━━━━━ 5s 126ms/step - loss: 0.0020
Epoch 81/100
38/38 ━━━━━━━━━━━━━━━━━━━━ 4s 107ms/step - loss: 0.0019
Epoch 82/100
38/38 ━━━━━━━━━━━━━━━━━━━━ 5s 121ms/step - loss: 0.0016
Epoch 83/100
38/38 ━━━━━━━━━━━━━━━━━━━━ 5s 120ms/step - loss: 0.0014
Epoch 84/100
38/38 ━━━━━━━━━━━━━━━━━━━━ 4s 104ms/step - loss: 0.0019
Epoch 85/100
38/38 ━━━━━━━━━━━━━━━━━━━━ 5s 136ms/step - loss: 0.0015
Epoch 86/100
38/38 ━━━━━━━━━━━━━━━━━━━━ 9s 108ms/step - loss: 0.0015
Epoch 87/100
38/38 ━━━━━━━━━━━━━━━━━━━━ 6s 144ms/step - loss: 0.0017
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Epoch 88/100
38/38 ━━━━━━━━━━━━━━━━━━━━ 9s 109ms/step - loss: 0.0018
Epoch 89/100
38/38 ━━━━━━━━━━━━━━━━━━━━ 5s 139ms/step - loss: 0.0019
Epoch 90/100
38/38 ━━━━━━━━━━━━━━━━━━━━ 9s 114ms/step - loss: 0.0015
Epoch 91/100
38/38 ━━━━━━━━━━━━━━━━━━━━ 5s 120ms/step - loss: 0.0015
Epoch 92/100
38/38 ━━━━━━━━━━━━━━━━━━━━ 5s 106ms/step - loss: 0.0017
Epoch 93/100
38/38 ━━━━━━━━━━━━━━━━━━━━ 5s 130ms/step - loss: 0.0014
Epoch 94/100
38/38 ━━━━━━━━━━━━━━━━━━━━ 4s 109ms/step - loss: 0.0015
Epoch 95/100
38/38 ━━━━━━━━━━━━━━━━━━━━ 4s 104ms/step - loss: 0.0016
Epoch 96/100
38/38 ━━━━━━━━━━━━━━━━━━━━ 5s 131ms/step - loss: 0.0016
Epoch 97/100
38/38 ━━━━━━━━━━━━━━━━━━━━ 4s 109ms/step - loss: 0.0015
Epoch 98/100
38/38 ━━━━━━━━━━━━━━━━━━━━ 5s 104ms/step - loss: 0.0015
Epoch 99/100
38/38 ━━━━━━━━━━━━━━━━━━━━ 5s 141ms/step - loss: 0.0015
Epoch 100/100
38/38 ━━━━━━━━━━━━━━━━━━━━ 9s 105ms/step - loss: 0.0015

<keras.src.callbacks.history.History at 0x7afdef505f60>
```

```python
#Part 3 - Making the predictions and visualising the results

#Getting the real stock price of 2017
dataset_test = pd.read_csv('/content/Google_Stock_Price_Test.csv')
real_stock_price = dataset_test.iloc[:, 1:2].values

#Getting the predicted stock price of 2017
dataset_total = pd.concat((dataset_train['Open'],
dataset_test['Open']), axis = 0)
inputs = dataset_total[len(dataset_total)-len(dataset_test) -
60: ].values
inputs = inputs.reshape(-1,1)
inputs = sc.transform(inputs)
X_test = []
for i in range(60, 80):
    X_test.append(inputs[i-60:i, 0])
X_test = np.array(X_test)
X_test = np.reshape(X_test, (X_test.shape[0], X_test.shape[1], 1))
predicted_stock_price = regressor.predict(X_test)
predicted_stock_price = sc.inverse_transform(predicted_stock_price)
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#Visualing the results
plt.plot(real_stock_price, color = 'red', label = 'Real Google Stock
Price')
plt.plot(predicted_stock_price, color = 'green', label = 'Predicted
Google Stock Price')
plt.title('Google Stock Price Prediction')
plt.xlabel('Year')
plt.ylabel('Google Stock Price')
plt.legend()
plt.show()
```